

Practice Exercise

Laplace Transform (CO1)

Find the Laplace transformation of the following

1. $e^{2t} + 4t^3 - \sin 2t \cos 3t$

2. $e^{-3t} \cosh 4t \cos 2t \sin 2t$

3. $\sin^5$

4. $te^{3t} \cos 2t \sin \sin 2t$

5. $\frac{\sin t \sin 5t}{t}$

6. $\frac{d}{dt} \left(\frac{\sin t}{t} \right)$

7. $\int_0^t e^{-t} \frac{(1 - \cos 2t)}{2t} du$

8. $t \cdot \operatorname{erf}(\sqrt{t})$

Evaluate the following integrals

1. $\int_0^\infty \frac{\sinh 2t \sin 3t}{e^{-t}} dt$

2. $\int_0^\infty \frac{t \sin t}{e^{-3t}} dt$

3. $\int_0^\infty \frac{\cos 6t - \cos 4t}{t} dt$

4. $\int_0^\infty \frac{1}{e^{2t}} \int_0^\infty \frac{e^{-u} \sin u}{u} du dt$

5. $\int_0^\infty e^{-3t} \cos^2 t dt$

6. $\int_0^\infty \cos 2t \sin 3t dt$

Inverse Laplace Transform (CO2)

Find inverse Laplace transform of the following using partial fraction.

1. $\frac{s+29}{(s^2+9)(s+4)}$

2. $\frac{s}{(s+1)^2(s^2+1)}$

Find inverse Laplace transform of the following using convolution theorem.

3. $\frac{s^2}{(s^2+1)(s^2+4)}$

4. $\frac{16}{(s+2)(s-2)}$

Find inverse Laplace transform of the following.

5. $\frac{1}{2} \log \frac{s-1}{s+1}$

6. $\frac{1}{s}$

6. Using Laplace transform solve following differential equations with the given conditions.

a. $(3D + 2)y = e^{3t}, y(0) = 1$

b. $(D^2 + 2D + 1)y = 3te^{3t}, y(0) = 4, y'(0) = 2$

Fourier Series (CO3)

1. Find the Fourier series for $f(x) = \frac{1}{4}(\pi - x)^2$ in $(0, 2\pi)$.

Hence deduce that $\frac{\pi^4}{90} = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \dots$

2. Find the Fourier series for $f(x) = \{x - \pi, \quad -\pi < x < 0 \quad \pi - x, \quad 0 < x < \pi,$

Hence deduce that $\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

3. Find the Fourier series for $f(x) = |\sin x|$ in $(-\pi, \pi)$.
4. Find the Fourier series for $f(x) = x \sin x$ in the interval $0 \leq x \leq 2\pi$
5. Obtain half range cosine series for $f(x) = x(\pi - x)$ in $(0, \pi)$.

$$\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots$$

6. Obtain half range sine series for $f(x) = (lx - x^2)$ in $(0, l)$.

$$\frac{\pi^6}{960} = \frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \frac{1}{7^6} + \dots$$

7. Find the complex form of Fourier series for $f(x) = \cos hx + \sin hx$ in $(-l, l)$.
8. Show that the set of functions $\sin x, \sin 2x, \sin 3x, \dots$ is orthogonal on $(0, \pi)$,
Hence construct a set of orthogonal functions.

Complex Variables (CO4)

1. Show that the $f(x) = e^{-x}(x \cos y + y \sin y)$ is Harmonic and find the corresponding analytic function.
2. Find the orthogonal trajectories of the family of curve given by $e^x \cos y - xy = c$.
3. Show that function $u(x, y)$ is harmonic and find the corresponding analytic function

$$f(z) = u + iv, \text{ where } u = \frac{1}{2} \log(x^2 + y^2).$$

4. Find the analytic function in terms of z if $u + v = \frac{x}{x^2 + y^2}$.
5. Show that the function $w = \log z$ is analytic except at $z = 0$.
6. Construct an analytic function $f(z) = u + iv$ by using Milne Thompson's method whose imaginary part is $v = e^{-x}(y \sin y + x \cos y)$.
7. Find the analytic function $f(z) = u + iv$ in terms of z , if $u - v = x^3 + x^2 - 3xy^2 - y^2 - 3x^2y + y^3 - 2xy$.
8. Find k such that $f(z) = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1} \frac{kx}{y}$ is analytic.

Statistical Technique (CO5)

1. Obtain the rank correlation coefficient from the following data

X	10	12	18	18	15	40
Y	12	18	25	25	50	25

2. In a contest for a search for Mr. Puerto Princesa City 2024, two judges gave their ratings to 8 candidates. Compute the coefficient of rank correlation.

Judge 1	97	96	94	89	88	87	84	84
Judge 2	93	96	97	94	91	89	88	84

3. Obtain the rank correlation coefficient from the following data

X	10	12	18	18	15	40
Y	12	18	25	25	50	25

4. The equation of two lines of regression are

$$2x - 9y = -6 \text{ and } x - 2y = -1.$$

Find (a) means of x and y

(b) the coefficients of correlation between x and y .

(c) the S.D. of y if $Var(x) = 9$

5. Fit a second-degree curve $Y = a + bx + cx^2$, to the following data.

X	0	1	2	3	4
Y	-4	-1	4	11	20

6. Fit a second degree parabolic curve to the following data and estimate the production in 1982

Year(X)	1974	1975	1976	1977	1978	1979	1980	1991
Production(Y) (in tons)	12	14	26	42	40	50	52	53

7. In a contest for a search for Mr. Puerto Princesa City 2024, two judges gave their ratings to 8 candidates. Compute the coefficient of rank correlation.

Judge 1	97	96	94	89	88	87	84	84
Judge 2	93	96	97	94	91	89	88	84

8. Calculate Spearman's coefficient of rank correlation from the following data

X	18	20	34	52	12
Y	39	23	35	18	46

Probability (CO6)

1. In a bolt factory, machines A, B and C manufacture respectively 25%, 35% and 40% of their total. Their output 5, 4 and 2 percent are defective bolts. A bolt is drawn at random from the product and is found to be defective. What is the probability that it was manufactured by machine B.

2. Find k, mean and the variance of X if the probability density of a

$$\text{random variable is given by } f(x) = \begin{cases} kxe^{-\frac{x}{3}}, & x > 0 \\ 0 & , x \leq 0 \end{cases}$$

3. Find the first two moments and M.G.F. of the RV X about origin whose p.m.f. is given by

X	-2	3	1
P(X = x)	1/3	1/2	1/6

4. Find the moment generating function of the following distribution

X	-2	3	1
P(X = x)	1/3	1/2	1/6

Hence find central moments. $\mu_1, \mu_2, \mu_3, \mu_4$.

5. Find the value of k, if the function

$$f(x) = \begin{cases} kx^2(1 - x^3) & , 0 \leq x \leq 1 \\ 0 & , \text{otherwise} \end{cases}$$

is a probability density function. Also find $P(0 \leq x \leq 1/2)$, mean, variance.

6. A continuous random variable X has the following probability law

$$f(x) = kx^2, 0 \leq x \leq 2$$

Find k and find the probability that (i) $(0.2 \leq x \leq 0.5)$

(ii) $x \geq 3/4$, given that $x \geq 1/2$.

7. 1% of women at age forty who participate in routine screening have cancer. 80% of women with cancer will get positive mammographies. 9.6% of women without cancer will also get positive mammographies. A woman in this age group had a positive mammography in a routine screening, what is the probability that she actually has cancer?

8. The daily consumption of electric power is a random variable x with probability density

$$\text{function } f(x) = \begin{cases} kxe^{-\frac{x}{3}}, & x > 0 \\ 0, & x < 0 \end{cases} \text{ Find the value of k, the expectation of x, and the}$$

probability that on a given day electric consumption is more than the expected value.